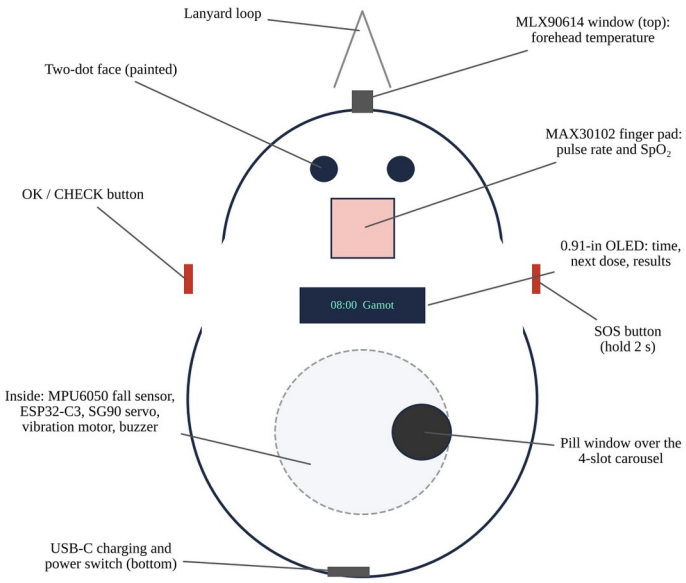
ENGINEERING NOTEBOOK

KALINGA

Development of a 3D-Printed, Baymax-Inspired Wearable Health Pendant



Front view, actual size about 60 × 83 × 39 mm, 3D-printed in white PLA.

The pendant to be built (front view).

Team: _____, _____, _____ Adviser: _____ School: _____

How to keep this notebook. Write every entry on the day the work is done. Sign and date each entry and have a witness sign. Do not erase or tear pages: cross out mistakes with one line. Paste printouts, photos, and screenshots with the date. Record what failed as carefully as what worked; judges look for the design process, not only the final product.

Entry 1: Problem and Design Criteria

Date: _____ Recorded by: _____ Witnessed by: _____

Problem. Older adults and people with chronic illness are often alone at home while their families work. Falls go unnoticed, doses are missed, and fevers rise unseen. Existing products each do one job and cost more together.

Goal. Build one low-cost wearable that checks vital signs, presents each dose on time, alerts the family, and shows everything in a phone app.

Criterion	Target	How it will be checked
Cost	Below PHP 2,500 for parts	Receipts (Appendix A.3)
Size and weight	Fits in a hand; about 60 × 83 × 39 mm; under 120 g	Ruler and kitchen scale
Temperature accuracy	Bias within ±0.3 °C after calibration	Bland–Altman vs clinic thermometer
SpO_2 accuracy	Arms ≤ 4% (ISO 80601-2-61)	Bland–Altman vs reference oximeter

Fall detection	Sensitivity and specificity ≥ 90%	Mat falls and daily activities
Alert time	Under 60 s to the app	Stopwatch and app time stamps
Doses	Correct slot every time; within 1 min of schedule	5-day dose log
Battery	At least 16 h per charge	Battery log
Ease of use	SUS above 68	Questionnaire

Entry 2: Background Research

Date: _____ Recorded by: _____ Witnessed by: _____

- Pulse oximetry: red and infrared light through the fingertip give SpO_2 and pulse rate (Allen, 2007; Jubran, 2015). Readings drop in accuracy with movement, cold fingers, nail polish, and darker skin.
- Infrared thermometry: the MLX90614 reads skin temperature without contact; forehead skin is cooler than the body core, so a calibration equation is needed (Fraden, 2016).
- Falls: free fall (< 0.5 g), then impact (> 2.5 g), then stillness is a simple, proven rule (Bourke et al., 2007).
- Medicine: about half of long-term patients do not take medicines as prescribed; timed reminders help (WHO, 2003).
- Data: the Firebase Realtime Database has a REST interface that a microcontroller can call over HTTPS and a web app can stream live (Google, n.d.). Health data must be protected (Republic Act No. 10173, 2012).

Sources read by the team and notes: _____

Entry 3: Ideas and Decision Matrix

Date: _____ Recorded by: _____ Witnessed by: _____

Each idea is scored 1 (poor) to 5 (best) on each criterion and multiplied by the weight. The scores below are the starting estimates; change them after your own tests and sign the decision.

Option for each function	Cost ×3	Size ×2	Ease to build ×2	Reliability ×3	Total
Dispensing: 4-slot carousel + SG90 servo	5	3	4	4	41
Dispensing: single lid with servo latch	5	4	4	3	40
Dispensing: sliding drawer + linear servo	3	4	2	3	30
Temperature: MLX90614 non-contact	3	4	5	4	39
Temperature: MAX30205 skin contact	4	5	3	3	37
Link: Wi-Fi + Firebase + web app	5	5	3	4	43
Link: Bluetooth to the older adult's phone	5	5	3	2	37
Link: GSM SMS module	3	2	3	3	28

Chosen design: carousel + SG90, MLX90614, Wi-Fi + Firebase. Team signatures: _____

Entry 4: Bill of Materials and Tools

Date: _____ Recorded by: _____ Witnessed by: _____

Item	Qty	Est. PHP
ESP32-C3 SuperMini	1	220
MLX90614 (GY-906) module	1	550
MAX30102 module	1	180
MPU6050 (GY-521) module	1	120
0.91-in OLED, 128 × 32, I²C	1	150
SG90 micro servo	1	120
LiPo 3.7 V 1000 mAh (503450)	1	220
TP4056 USB-C charger with protection	1	45
Coin vibration motor, 3 V active buzzer	1 set	50
AO3400, S8050, 1N4148, resistors	1 set	30
Tactile buttons with caps, slide switch	1 set	35
Perfboard, wires, headers	1 lot	80
PLA filament or print service (about 60 g)	1 lot	250
Black acrylic paint, sandpaper	1 lot	40
Breakaway lanyard, M2 screws	1 set	60
TOTAL (estimated)		**2,150**

Tools. 3D printer (FDM, 0.4-mm nozzle) or a print shop; slicer (UltiMaker Cura, PrusaSlicer, or Bambu Studio); soldering iron, solder, flux; wire stripper; multimeter; hot-glue gun; small files and sandpaper (220 and 400); flush cutters; laptop with Arduino IDE 2; USB-C cable; placebo candy tablets for testing.

Entry 5: CAD Design of the Shell

Date: _____ Recorded by: _____ Witnessed by: _____

The shell is an egg-shaped body (two spheres joined and flattened) split into a front and a back shell, plus a 36-mm pill carousel. The model is a Python script, so every size can be changed and the STL files regenerated:

pip install manifold3d trimesh then *python cad/kalinga_pendant.py*

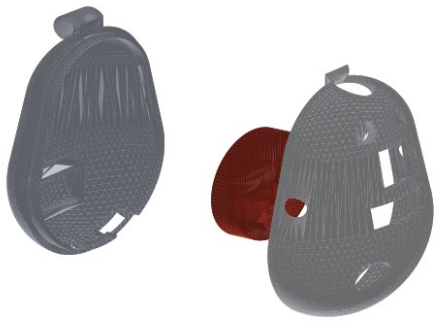
Feature	Size (mm)	Why
Wall thickness	2.0	Strong enough for PLA; light
Overall	60 × 83 × 39	Fits the SG90 servo and carousel
OLED window	24 × 8	0.91-in OLED active area 22.4 × 5.6
Finger pad window	13 × 13	MAX30102 module
Thermometer window (top)	Ø 9.8	MLX90614 TO-39 can, Ø 9.2
Pill window	Ø 11	Pocket Ø 9.6 under it
Carousel	Ø 36 × 7, 4 pockets	Pockets at 45°, 90°, 135°, 180°; solid at 0° = closed
Button holes	Ø 6.6	6 × 6 mm tactile button caps
USB-C and switch slots	10 × 4, 9 × 4	TP4056 port, slide switch
Lip	3 deep, 1.2 thick	Locates the front shell on the back shell

Front shell (face)



Front shell rendered from front_shell.stl.

Exploded view: back shell, carousel, front shell



Exploded view: back shell with servo cradle, pill carousel, and front shell.

Changes made to the CAD and why: _____

Entry 6: 3D Printing

Date: _____ Recorded by: _____ Witnessed by: _____

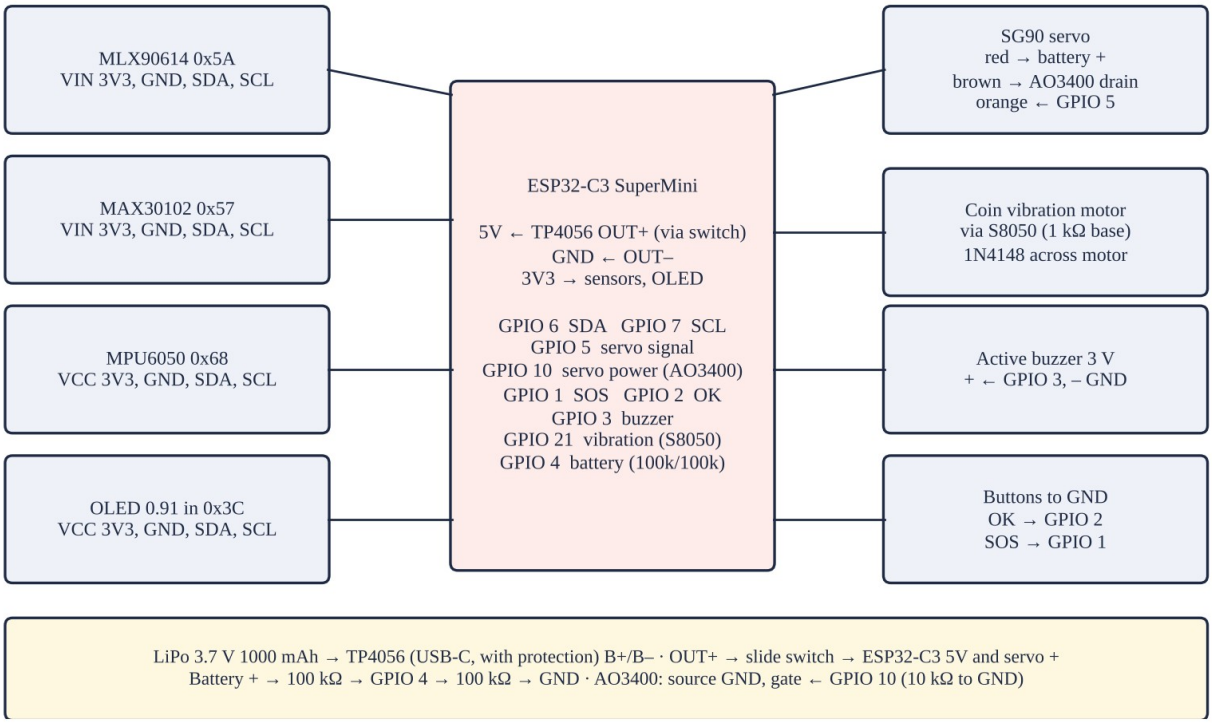
Setting	Shells	Carousel
Material	White PLA	White or red PLA
Nozzle / layer height	0.4 mm / 0.2 mm	0.4 mm / 0.2 mm
Walls / top and bottom layers	3 / 4	3 / 4
Infill	20% gyroid	30% gyroid
Nozzle / bed temperature	205 °C / 60 °C	205 °C / 60 °C
Orientation	Flat split edge on the bed (open side down)	Flat, pockets facing up
Supports	Tree supports, everywhere; 5-mm brim	None
Estimated time and filament (from slicer)	_____ h, _____ g	_____ h, _____ g

1. Print the carousel first as a test: check that a placebo tablet fits each pocket and that the SG90 horn sits flat in its recess.
2. Print the back shell, then the front shell. Watch the first layer; stop and level the bed if lines do not stick.
3. Remove supports with flush cutters and pliers. Sand edges with 220 then 400 grit.
4. Dry-fit the two shells: the lip must slide in with light pressure. If tight, sand the lip; if loose, add one layer of tape.
5. Paint the two eye recesses with black acrylic; let dry for 1 hour. Keep the face design original (two dots, no copied logos) so it can be filed as an Industrial Design.
6. Weigh and measure each part and paste photos here.

Part	Print date	Time (h)	Mass (g)	Problems seen	Fix

Entry 7: Electronics Assembly

Date: _____ Recorded by: _____ Witnessed by: _____



Wiring diagram.

Part	Connects to (ESP32-C3)
MLX90614, MAX30102, MPU6050, OLED	GPIO 6 / 7 — I²C at 100 kHz, 3.3 V
SG90 servo	GPIO 5 / GPIO 10 — AO3400 switches the servo ground
OK and SOS buttons	GPIO 2 / GPIO 1 — Internal pull-ups
Buzzer; vibration motor	GPIO 3; GPIO 21 — 1 kΩ base resistor; 1N4148
Battery divider	GPIO 4 — Reads 0–4.2 V
TP4056 OUT+ / OUT–	5V / GND — LiPo on B+ / B–

1. Solder headers on each module. Test each module alone with the ESP32-C3 and an I²C scan: expect 0x5A (MLX90614), 0x57 (MAX30102), 0x68 (MPU6050), and 0x3C (OLED).
2. Build the small parts on perfboard: S8050 with 1 kΩ base resistor and 1N4148 across the vibration motor; AO3400 with 10 kΩ gate-to-ground resistor for the servo; 100 kΩ/100 kΩ battery divider to GPIO 4.
3. LiPo safety: connect the battery to the TP4056 B+ and B– only, with the correct polarity, and never short, bend, or puncture it. Charge only under supervision.
4. Wire TP4056 OUT+ through the slide switch to the ESP32-C3 5V pin and the servo red wire; OUT– to GND.
5. Measure with the multimeter before powering: battery 3.7–4.2 V; no short between 5V and GND.
6. Record each test below.

Module	Test	Expected	Result	OK?

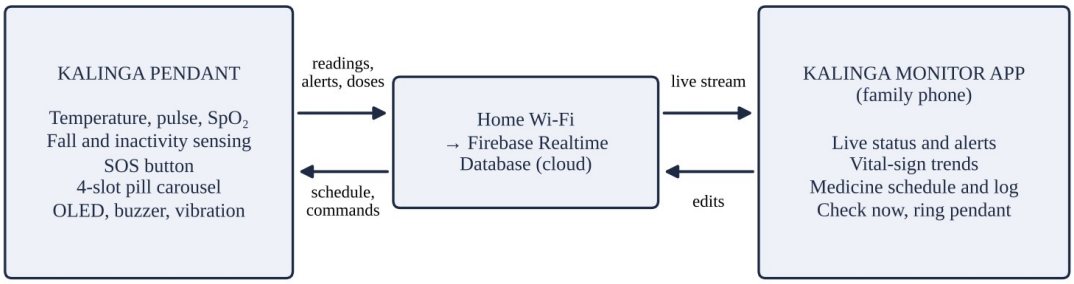
Entry 8: Programming the Pendant

Date: _____ Recorded by: _____ Witnessed by: _____

1. Install Arduino IDE 2 and “esp32 by Espressif Systems” in Boards Manager.
2. Install libraries: Adafruit MLX90614, Adafruit SSD1306, Adafruit GFX, SparkFun MAX3010x, ESP32Servo, ArduinoJson.
3. Open firmware/kalinga_pendant/kalinga_pendant.ino. Type the Wi-Fi name and password, the Firebase Web API key and database URL, the pendant's account email and password, and the patient id.
4. Board: ESP32C3 Dev Module; USB CDC On Boot: Enabled. Upload. Open Serial Monitor at 115200.
5. Check: the OLED shows the time and battery; the carousel moves to the closed position; holding SOS for 2 s sends an SOS alert.

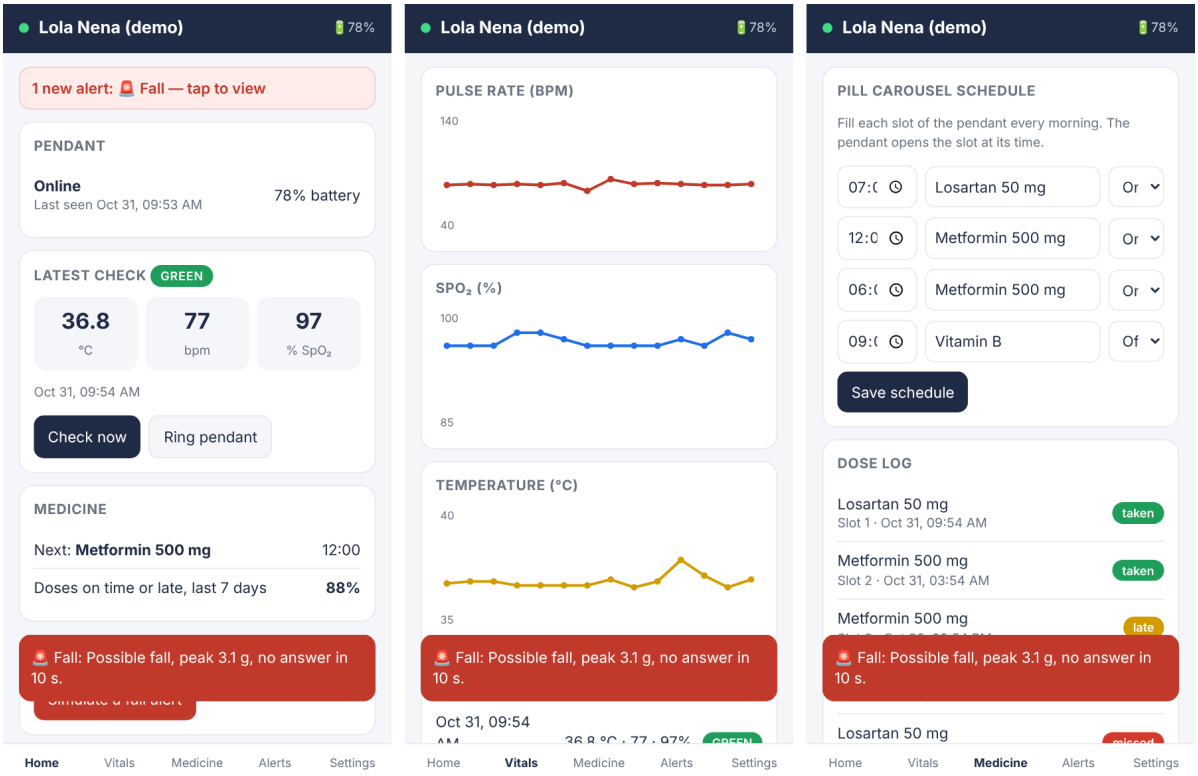
Entry 9: Firebase and the KALINGA Monitor App

Date: _____ Recorded by: _____ Witnessed by: _____



How the pendant, the database, and the app connect.

1. Go to console.firebase.google.com, create a project, and add a Web app. Copy the Web API key.
2. Authentication → Sign-in method → enable Email/Password. Add one user for the pendant and one for each family member. Copy each user's UID.
3. Realtime Database → Create (location: Singapore, asia-southeast1) → start in locked mode. Copy the database URL.
4. Rules: paste app/database.rules.json and publish.
5. Data: add members → <patient id> → <each UID> = true (for the pendant and every family member).
6. Host the app folder on GitHub Pages or Firebase Hosting (HTTPS is needed to install it). Open it on a phone, add it to the home screen, enter the key, URL, patient id, email, and password, and sign in.
7. Use “Try the demo” to show the app to judges without the pendant.



KALINGA Monitor screens (demo mode).

Entry 10: Final Assembly

Date: _____ Recorded by: _____ Witnessed by: _____

1. Set the servo to the closed position (power on once; the program moves it to 0°) before fitting the horn and carousel, so the solid sector sits under the window.
2. Fix the servo in the back-shell cradle with a drop of hot glue; screw the single-arm horn into the carousel recess.
3. Hot-glue the OLED behind its window, the MAX30102 behind the finger window (sensor flush with the surface), and the MLX90614 can in the top hole.
4. Place the ESP32-C3, MPU6050 (flat, near the center), battery, and TP4056 with double-sided foam tape; keep the USB-C port at its slot.
5. Fit the buttons and caps, close the shells, and add two M2 screws or a drop of glue at the bottom.
6. Thread the breakaway lanyard. Fill the four pockets with placebo tablets and run one dose cycle.

Entry 11: Calibration

Date: _____ Recorded by: _____ Witnessed by: _____

1. Bench test of the thermometer with a black-painted can in a water bath at 35.0, 36.0, 37.0, 37.5, 38.0, and 40.0 °C against a probe thermometer.
2. Calibration set (10 volunteers, after IRB approval): TEMP_A = SLOPE(reference, raw), TEMP_B = INTERCEPT(reference, raw); PR and SpO_2 offsets = mean of (reference − pendant).
3. Fall thresholds: 10 mat falls and 10 daily activities by the researchers; adjust FREE_FALL_G and IMPACT_G, then keep them fixed.

Target or volunteer	Reference	Pendant raw	Pendant calibrated	Difference

Entry 12: Test Records

Date: _____ Recorded by: _____ Witnessed by: _____

Falls and daily activities

Trial	Event	Detected?	Cancelled?	Peak g	Alert time (s)

Doses

Date	Slot	Scheduled	Opened at	Recorded as	Observer says

Battery

Charge no.	Full at	Empty at	Hours	Notes

Entry 13: Problems and Iterations

Date: _____ Recorded by: _____ Witnessed by: _____

Write each problem the day it happens, what caused it, what you changed, and what happened after.

Date	Problem	Cause	Change made	Result

Entry 14: Safety Checklist and Reflection

Date: _____ Recorded by: _____ Witnessed by: _____

- Adult supervision for the 3D printer, soldering, and LiPo charging.
- IRB approval and signed consent before any test with people; placebo tablets only.
- Falls simulated only by the researchers on 20-cm mats with pads.
- Breakaway lanyard so the cord cannot strangle the wearer.
- Firebase locked by rules; coded patient id; data deleted after the fair.

What worked: _____

What we would change next: _____